Energy Research & Social Science 130 (2025) 104415

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Original research article

Can renewable energy zones become regional equity zones? Policy coherence and social equity in the Electricity Infrastructure Roadmap in New South Wales, Australia

Jonathan Pickering [a,*], Pierrick Chalaye [b]

[a] Canberra School of Government, Building 24, University Drive South, University of Canberra, Bruce, ACT 2617, Australia
[b] CERTOP UMR 5044, Maison de la Recherche, Université Toulouse – Jean Jaurès, 5, allée Antonio Machado, F-31058 TOULOUSE CEDEX 9, France

ARTICLE INFO

Keywords:
Policy coherence
Energy transition
Renewable energy zones
Social equity
Community engagement
Benefit-sharing
Australia

ABSTRACT

Governments across a range of countries are establishing renewable energy zones to accelerate and coordinate the construction of large-scale solar, wind and battery infrastructure and associated transmission lines, mainly in regions outside metropolitan areas. Renewable energy zones intensify the geographical concentration of both the benefits and impacts for communities living near this infrastructure, raising both opportunities and challenges for policymaking. This article applies theories of policy coherence to ask: (a) what synergies and tensions arise between renewable energy and social equity for regional communities when establishing and implementing renewable energy zones, and (b) what political factors drive synergies and tensions between these policy objectives and how could they be managed more effectively? We present findings from a case study of the Electricity Infrastructure Roadmap in New South Wales, Australia's most populous state, drawing on documentary analysis, interviews and focus groups. We identify several drivers of incoherence, including political polarisation over energy policy, limited inter-agency coordination and scale mismatches that disincentivise community energy initiatives. Drivers of coherence include cross-party dialogue, strengthening institutional capacity and incentives for project developers to invest in community engagement and benefit-sharing. Our analysis yields lessons for the rollout of renewable energy zones internationally, while also demonstrating the value of a policy coherence lens for advancing equity and justice together with other policy objectives.

1. Introduction

Governments across a range of countries are seeking to coordinate the buildout of new large-scale renewable energy infrastructure through designated renewable energy zones (REZs) or similar frameworks, including in Australia, India, the Philippines, South Africa and the United States [1]. While there has been extensive interest in the opportunities that renewable energy offers to decentralise energy production away from fossil fuel power plants, REZs represent a partial recentralisation of power generation, thus interacting with other centripetal dynamics such as competitive auctions that favour utility-scale renewable energy projects [2]. As one report notes, "REZs are the equivalent of modern-day power stations" [3], albeit spread over many square kilometres.

Large-scale renewable energy projects offer a range of potential benefits for regional[1] communities – including employment, landholder payments and community benefit-sharing schemes – but they can also yield a variety of adverse impacts, including disruption of visual amenity and land use, and economic volatility [4]. The design of REZs, which makes it more likely that a given area will host multiple projects, "concentrates both the benefits and the impacts for local communities" (INT12N; see Appendix A1 for details of interview focus/group participants). While this may unlock opportunities that are harder to grasp with more dispersed projects (e.g. pooled benefit-sharing schemes, regional training initiatives and coordinated impact management), it also increases the challenge of dealing with adverse impacts and the risk of community opposition. Significant opposition of the kind increasingly in evidence in Australia [5] and elsewhere [6,7] can in turn impede the

* Corresponding author.
E-mail address: jonathan.pickering@canberra.edu.au (J. Pickering).

[1] In Australia, the term "regional" is often used as shorthand for areas outside major capital cities, including smaller cities and towns, rural communities and remote areas. We follow the same practice in this article.

https://doi.org/10.1016/j.erss.2025.104415
Received 14 April 2025; Received in revised form 13 October 2025; Accepted 20 October 2025
Available online 28 October 2025
2214-6296/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

pace and scope of the energy transition. Accordingly, understanding how the establishment and implementation of REZs affect the pursuit of other policy objectives (including social equity and the wellbeing of rural communities) is an important step towards robust, fair and publicly supported energy policy [8].

This article asks: (a) what synergies and tensions arise between renewable energy and social equity in the establishment and early implementation of REZs and how are those synergies and tensions being managed so far, and (b) what political factors drive synergies and tensions between these policy objectives and how could they be managed more effectively? We explore these questions through the lens of policy coherence, which can be understood as an approach to policy-making that minimises conflicts among policy objectives and maximises synergies among them [9,10]. A coherent approach in this context would mean that the expansion of renewable energy simultaneously advances social equity (thus enabling REZs to become "regional equity zones" too), while an incoherent approach would mean that renewable energy expansion exacerbates inequity or fails to alleviate it. We present a case study of renewable energy expansion in New South Wales (NSW), which is Australia's most populous state and a first mover on REZs nationally [11]. Our analysis focuses on early implementation of the NSW Electricity Infrastructure Roadmap ("the Roadmap"), a policy framework launched by the NSW Government in 2020 that aims to expand the state's renewable energy capacity, primarily through the construction of REZs, while also pursuing a range of equity-related objectives.

Section 2 sets out the analytical lens of policy coherence that we employ in the article. Section 3 provides an overview of the case study context and methods. Section 4 identifies areas of coherence and incoherence in the early implementation of the NSW Roadmap. Our analysis draws on government policies, reviews and media reports, as well as interviews and focus groups involving forty participants. We identify several areas of incoherence, including limited progress in ensuring that First Nations people, low-income groups and rural communities can harness the benefits of the energy transition despite a range of regulatory measures intended to facilitate this, as well as some areas of coherence, including enhanced benefit-sharing mechanisms and bolstering institutional capacity for community engagement. In section 5 we employ a recently developed framework for analysing political drivers of coherence and incoherence spanning ideas, institutions and interests [10]. We identify several drivers of incoherence, including political polarisation over energy policy, limited inter-agency coordination and scale mismatches that disincentivise community-owned energy initiatives. Drivers of coherence include cross-party dialogue, strengthening institutional capacity, and incentives for project developers to invest in community engagement and benefit-sharing. Section 6 then discusses the findings and draws out a range of policy implications.

Through this analysis, the article aims both to advance theory, by demonstrating the usefulness of a policy coherence lens for revealing, explaining and managing synergies and trade-offs in energy policy, and to contribute to the emerging body of empirical evidence on the socio-economic consequences of REZs. More broadly, the article also aims to further scholarly understanding of opportunities and constraints for advancing energy justice and social equity [12], as well as the politics of scale around energy infrastructure [13,14].

2. Analytical framework: policy coherence and social equity in energy transitions

2.1. A policy coherence lens on energy transitions and social equity

The need for energy policy to manage trade-offs among multiple policy objectives is often conceptualised as a "trilemma" that requires balancing the security, sustainability and affordability of energy supply [15].[2] While this concept is useful insofar as it highlights the challenge of balancing multiple priorities in energy policy, conventional framings of the energy trilemma have been criticised for downplaying other important values or objectives, including procedural and recognitional justice, and other facets of wellbeing that are affected by energy, including community cohesion, autonomy and attachment to place (see e.g. [16]). Thus, there is value in exploring more comprehensive and flexible frameworks for analysing trade-offs and synergies. A policy coherence perspective on energy policy may also complement an energy conflict lens [6] while offering additional insights, as it can reveal areas of policy incoherence that may produce adverse effects even if they do not manifest in outright conflict, while also highlighting areas of synergy as well as areas of tension or conflict.

Policy coherence has become increasingly cited as both a desirable feature of contemporary governance and as an analytical lens for analysing trade-offs and synergies. The United Nations Sustainable Development Goals, for example, include a target calling upon states to "Enhance policy coherence for sustainable development" (target 17.14). Policy coherence for sustainable development can be defined as an "approach and policy tool to systematically integrate the social, economic and environmental dimensions of sustainable development at all stages of policy making" [17]. A growing body of literature has begun to apply a policy coherence lens to the context of energy, but much of this work focuses on the European Union [8] with limited coverage of other regions. Moreover, there has been limited systematic investigation of the factors that enable or constrain policy coherence in energy policy, as we elaborate in section 2.2 below.

At various stages of the policy process from agenda-setting and policy formulation to implementation, (in)coherence may arise (a) between different policy objectives (*horizontal* [in]coherence, e.g. between renewable energy expansion and local employment) at one level of governance, or (b) in the pursuit of a single objective by actors or institutions at different levels (*vertical* [in]coherence, e.g. where federal and state policies on energy overlap, clash or reinforce one another) [8]. Horizontal and vertical dimensions may also interact (e.g. where a national government's pursuit of renewable energy supports or obstructs the ability of a subnational government to encourage local employment). We mainly address horizontal coherence at the subnational/state level between two sets of policy objectives: (a) climate change mitigation through the expansion of renewable energy ("energy objectives"); and (b) the pursuit of social equity ("equity objectives").[3]

We refer to social equity (or in brief "equity") in the broad sense of "fair treatment and access to opportunities for all" [18], accounting for differentiated needs and historical legacies/injustices in the circumstances of both individuals and social groups [19]. We understand equity in both procedural and substantive terms, comprising both (a) the ability of individuals and social groups (particularly marginalised or vulnerable groups) to be recognised and participate in decision-making as well as (b) the fair distribution of the social and economic burdens and benefits associated with the energy transition among individuals and social groups [20].[4] Building on key equity concerns identified in previous research on energy justice [21] and on REZs [4] as well as an inductive approach informed by initial interviews for this project, our

[2] The latter objective is sometimes characterised as "equity" but often measured in relatively narrow terms of affordability and access (e.g. [15]).

[3] We briefly touch on interactions between horizontal and vertical coherence in section 5.3.2.

[4] Both dimensions align with common dimensions of energy justice [21], while the procedural aspect meshes with concerns of energy democracy [20]. However, space precludes a detailed discussion of the relationship between social equity and these related terms, or of other dimensions of equity and justice (e.g. recognition).

analysis focuses on procedural and distributive equity for three social groups: (i) rural communities in REZs, and within them (ii) low-income and marginalised groups, and (iii) First Nations people (Aboriginal and Torres Strait Islander people). For reasons of space we exclude detailed discussion of equity for people outside REZs, while acknowledging that both the upfront costs of REZ construction and the longer-term savings from the shift to renewable energy will have significant implications for consumers across the electricity network via their effects on electricity prices (INT19G; INT20B; INT31N) [22].

2.2. Drivers of coherence and incoherence

The limited literature on drivers of policy (in)coherence has tended to focus on the role of institutional factors such as interdepartmental coordination or mainstreaming and policy integration strategies [10,23]. However, a narrow institutional or technocratic focus may overlook the importance of political factors in facilitating coherence or obstructing it. Recent research has sought to gain greater conceptual clarity about political drivers of (in)coherence by employing a typology that distinguishes three categories of drivers: ideas, institutions and interests [10]. *Ideas* refer to "claims about descriptions of the world, causal relationships, or the normative legitimacy of certain actors" [24]. *Institutions* refer to "a cluster of rights, rules, and decision making procedures that gives rise to a social practice, assigns roles to participants in the practice, and guides interactions among occupants of these roles" [25]. *Interests* are understood as "material or advocacy considerations of specific sectors and actors related to pursuing their objectives" [10]. While existing literature has recognised the role of these categories in influencing the *stringency* of energy policy (see e.g. [26] on ideas and [27] on institutions and interests), less work has been done to date on how these categories may drive *(in)coherence* in energy policy. Strambo et al. [28], for example, contend that factors such as framing (an ideational factor), political will and market conditions (interest-based factors) may influence coherence between energy security and climate policy objectives, but their analysis focuses primarily on measuring coherence rather than investigating its causes. Section 5 applies the three-part typology outlined above to the case study context.

3. Case study overview and methods

3.1. Case study selection and policy context

3.1.1. Renewable energy and socioeconomic inequity in Australia

Australia serves as a useful case study for understanding the socio-economic dynamics of renewable energy zones as it is one of the countries, along with the United States, that has seen the greatest level of policy interest and activity in REZ models [1].[5] Australia is one of the world's largest exporters of fossil fuels. Regarding the three groups outlined in section 2.1 on which our equity analysis focuses: (i) on average, people living in rural communities in Australia have lower incomes and lower levels of social support than people living in metropolitan areas [30,31]; (ii) Australia has a relatively low level of socio-economic inequality by world standards [32], but inequality remains a major issue; and (iii) First Nations people continue to experience poorer health and well-being and higher levels of energy insecurity than non-Indigenous Australians while grappling with legacies of dispossession and intergenerational trauma [29,33,34].

In Australia, responsibility for energy and climate policy is shared primarily between state and federal governments, with local governments having a subsidiary role. To accelerate progress towards its goal of achieving net zero greenhouse gas emissions by 2050, the Federal Government aims to generate 82 % of the country's electricity from renewable sources by 2030 [35]. The eight states and territories (including New South Wales) have an associated range of short-term climate and renewable energy targets.

While electricity grids have largely developed at the state level, electricity policy is now an area of intense interaction between the Federal Government and the states. Australia's National Electricity Market (NEM) covers all states and territories except the Northern Territory and Western Australia. The NEM is regulated through a co-ordinated set of state and territory legislation and policy (notably the National Electricity Law and National Electricity Rules) as well as institutions such as the Australian Energy Market Operator (AEMO), which has a key role in coordinating the development of REZs across the NEM, including via its biennial Integrated System Plans [36]. All jurisdictions participating in the NEM are establishing REZs, supported by state-specific legislation and policy. Federal-state coordination on energy is facilitated by intergovernmental bodies such as the Energy and Climate Change Ministerial Council.

3.1.2. The New South Wales Electricity Infrastructure Roadmap

Of Australia's eight subnational jurisdictions (states and territories), NSW has the largest population (including the largest Aboriginal and Torres Strait Islander population) and the largest economy [37]. Most of the state's electricity infrastructure – with the exception of regional distribution networks – has been partially or wholly privatised since 2010 ([38]). Despite having the smallest share of renewable energy generation after Queensland, and the largest total energy consumption in the country [39], NSW generates the largest amount of renewable energy and hosts the largest number of large-scale renewable energy projects [40].

In 2020 the NSW Government launched a new 20-year policy framework for renewable energy expansion (the NSW Electricity Infrastructure Roadmap; hereafter "the Roadmap") – and associated legislation (the Electricity Infrastructure Investment [EII] Act 2020). The primary purpose of the Roadmap is to drive renewable energy expansion – particularly to avert the risk of energy shortages following recent and impending closures of ageing coal-fired power plants – and thereby accelerate progress towards the state's greenhouse gas emissions reduction targets. This is to be achieved primarily through underwriting of new large-scale solar and wind generation, transmission and energy storage [11]. While pre-existing planning legislation and policy continue to regulate the construction of renewable energy infrastructure across the state (including outside REZ boundaries), the EII Act is a distinctive development in that it establishes a legislative basis for constructing REZs in designated areas of the state. Developers (or "proponents") are incentivised to propose projects in REZs through the opportunity to tender for rights to access the grid and for Long Term Energy Service Agreements (LTESAs), which aim to provide minimum revenue streams for investors while protecting consumers from excessively high costs [41].

The EII Act's first objective ("to improve the affordability, reliability, security and sustainability of electricity supply") recalls the energy trilemma [38], but this is only one of eleven objectives. Notably, the Act contains several objectives relating to distributive equity, including promoting local education, training, industry and jobs (section 3(1)). While the Act includes objectives aligned with procedural equity to "foster local community support" for investment in renewables (section 3(1)(d)) and to "promote consultation and negotiation" with Aboriginal landowners (section 3(2)(b), it does not foreground broader community engagement requirements [42]. The government-owned Energy Corporation of NSW (EnergyCo) is tasked with coordinating the overall rollout of REZs in NSW, of which five have been declared as of mid-

[5] A key driver for the introduction of REZs in Australia was an independent review of the electricity market in 2017 led by Chief Scientist Alan Finkel, which drew on the example of Competitive Renewable Energy Zones in Texas [29].

2025.[6]

Much of the present analysis focuses on the Central-West Orana REZ (CWO REZ), which was the first to be declared in 2021. This REZ covers an area of roughly 20,000 km^2 on the lands of the Wiradjuri, Wailwan and Kamilaroi people, surrounding several regional centres including Dubbo and Mudgee. The network capacity of this REZ was initially estimated at 3 GW, and in 2023 was doubled to 6 GW due to increasing demand for electricity in the state. During the case study period (2021−2023), the CWO REZ was still in its early stages of development, with much work focusing on initial community engagement, scoping of individual infrastructure projects and the first rounds of tenders for LTESAs.[7]

3.2. Methods

This article is based on a mixed-methods analysis comprising documentary analysis, focus groups, interviews and event observation. The documentary analysis draws on climate, energy and equity policies at the federal and NSW levels, as well as a range of academic literature, think tank and NGO reports, press articles and stakeholder submissions to policy processes. We also held thirty interviews and two focus groups between February 2021 and February 2023 with forty participants, principally federal, state/territory and local government officials, non-governmental organisations, First Nations organisations, universities and think tanks, and renewable energy project developers. Appendix A1 provides further details about interview participants. Appendix A2 includes a template interview script. The case study included a short period of fieldwork in the Mudgee region within the CWO REZ in February 2023, and participant observation in an end-of-year briefing organised by EnergyCo (online, December 2022) and a stakeholder consultation on the First Nations Clean Energy Strategy (Canberra, August 2023). Finally, a multistakeholder concluding workshop held in Sydney in September 2023 informed this article and a more detailed project report [43].

The analysis mainly investigates the earlier stages of REZ implementation from 2021 to 2023, although subsequent developments (up to July 2025) are noted where relevant, particularly if they corroborate or diverge from findings from the earlier period. Given that many projects were yet to be constructed by the end of 2023, much of the evidence presented in this article is based on participants' first-hand experiences of community engagement and interactions with developers, and to a lesser extent on attitudes towards pre-existing or prospective infrastructure, with less emphasis on participants' experiences of the construction process itself.[8] Studying this early period can still yield vital insights as it covers a crucial phase in which relations of trust or mistrust as well as institutional dynamics are established that may have crucial implications for later phases of implementation [45].

Policy documents, interview and focus group transcripts and observational notes were coded thematically using NVivo data analysis software according to policy objectives and categories and drivers of policy (in)coherence based on the framework set out in section 2. Areas and drivers of (in)coherence discussed in sections 4 and 5 were selected based on frequency of coding and feedback from interviewees and participants about which drivers were most important. In the interest of balancing analytical breadth and depth within space constraints, treatment of some areas and drivers is inevitably brief.

[6] EnergyCo is a statutory body that was established in the 1980s but had been more or less dormant until Roadmap implementation began [38].

[7] LTESAs for two large projects in the CWO REZ (the Stubbo Solar Farm and the Uungula Wind Farm - both 400 MW installations) were awarded in 2023. Both projects are under construction as of mid-2025.

[8] Community attitudes towards renewable energy infrastructure may change before and after construction, sometimes becoming more positive [44].

4. Policy coherence in early implementation of the New South Wales Electricity Infrastructure Roadmap

This section assesses evidence of (in)coherence between energy and equity objectives in the initial phase of implementing the NSW Electricity Infrastructure Roadmap. The analysis is differentiated according to the two dimensions of procedural and substantive equity set out in section 2.1 above.

4.1. Procedural equity

Our analysis of procedural equity spans opportunities for authentic and inclusive public engagement at three stages of the policy process: the leadup to establishing REZs (section 4.1.1), during the early implementation process (section 4.1.2) and on individual projects (4.1.3). We also highlight issues of coherence relating to the participation of particular social groups (4.1.4), including First Nations people (4.1.5). Initial policy implementation is *coherent* if efforts to expand renewable energy through the Roadmap tend to enhance procedural equity, and *incoherent* if those efforts either undermine procedural equity or fail to address existing procedural inequities.

4.1.1. Establishment of Renewable Energy Zones

Decisions about where to site REZs have important flow-on effects for the communities within them (see section 4.2 below). The EII Act requires the Minister to have considered "the views of the local community" in the REZ before declaring a zone (EII Act, section 19(4)(b)(iii)). "Investor and community support" was one of five criteria used to identify candidate REZs, in addition to factors such as potential to generate renewable energy, proximity to existing and proposed transmission infrastructure, and implications for agricultural land use and biodiversity conservation [46]. Candidate REZs were largely identified through a geospatial mapping exercise, and extent of community support was primarily derived from a separate regional planning process rather than from a tailored consultation strategy. In the CWO REZ, many community members were initially unclear about why their area had been chosen as a REZ (INT12N, INT20B, INT24N; see also [47], p.30). One local government official commented that the council was "blindsided" by the announcement and that the implications of the REZ were not explained in terms that would enable the council to develop an informed position on it (INT26G; see also [4]).

4.1.2. Community outreach and engagement across the REZ

Numerous respondents stated that community outreach and awareness raising in the Central-West Orana REZ had been limited in the early stages of the Roadmap (roughly late 2020 through to early 2022) (INT12N, INT17N, INT26G). They pointed out that these efforts had increased over time, partly because community groups and councils had raised concerns about lack of engagement but also because EnergyCo's staffing and organisational capacity had built up and COVID-19 restrictions had eased (INT12N, INT26G).

EnergyCo conducted a range of online and in-person community outreach events and surveys [48], opened an office in Dubbo, and established community reference groups for several REZs involving representatives from farming, environmental and business organisations and the broader community. However, some interviewees expressed concern about the accessibility of timely and appropriate information about the CWO REZ (INT12N, INT23N). As the Roadmap has evolved, numerous respondents cited the onset of consultation fatigue, leading to disengagement and mistrust (INT14B, INT17N, INT24N, INT26G, [49]).

While engagement under the Roadmap involved a range of different avenues for participation, most of them appeared to emphasise consultation rather than more intensive forms of participation such as co-

design/co-production or deliberation (INT19G, [42]). Two notable exceptions in NSW have been (i) the involvement of First Nations groups in co-designing the First Nations guidelines[9]; and (ii) a community-led participatory initiative in the town of Hay to identify priorities for their involvement in the South-West REZ [51].

4.1.3. Consultation and negotiations on individual projects

Whereas broader community outreach across the REZs is coordinated by EnergyCo, consultation on individual projects is usually led by project developers. Consultation both before and after the Roadmap's establishment was reported to be highly variable, with some developers being far more thorough and inclusive than others (INT24N). Participants also noted a range of concerns about consultation and negotiation processes, including: insufficient consultation with landholders and neighbours at an early stage (INT17N); inadequate information about the impacts of proposed projects (INT23N) or conversely information overload (INT25N); and divisive strategies used by some developers to secure consent for hosting projects (INT23N; see also [49]).

Several respondents mentioned the cumulative impacts of multiple projects as a major driver of community opposition and an impediment to community engagement:

> Some projects that did have social licence that were […] early movers now are finding that they've got community opposition because all of a sudden there's another project next door to their project, and […] the community that were quite happy to have this one wind farm now find themselves surrounded by wind farms or solar farms, and […] they're not happy.
>
> (INT14B; corroborated by INT15B, INT24N)

In their environmental impact assessment, developers are required to show how they will address cumulative impacts of multiple projects (INT22B), but some participants considered it important to take into account other stressors that their communities have experienced in recent years, including drought, bushfires, COVID-19, flooding, mice plagues and controversies over other major developments (INT25N, [47]).

4.1.4. Participation of different social groups

Respondents acknowledged the difficulty of engaging underrepresented or harder-to-reach social groups, including lower-income households. Several obstacles were identified, including limited internet access, feeling intimidated by the prospect of going to a community consultation meeting (INT20B), limited education inhibiting people's ability to understand information or advocate for themselves (INT09B, INT25N), and inappropriate timing of consultation events, e.g. during work hours (INT26G). Developers frequently contact local organisations that involve a wide cross-section of the community and engage locally based consultants for outreach, but even with wide-ranging publicity of events, turnout is often small (INT20B, INT26G).

The dynamics of consultation meetings can also reinforce existing inequalities of participation. For example, people in lower-income groups may be less confident in raising concerns than wealthier landowners, and settings such as town hall meetings – which often have an adversarial atmosphere – may discourage women's participation (FG02R, [52]).

4.1.5. Involvement of First Nations people

Respondents reported mixed views about the extent of engagement with First Nations people in implementing the NSW Roadmap. One respondent commented that there was little consideration of Aboriginal people's interests when the location of the REZs was decided (INT18R). Subsequently, EnergyCo worked with First Nations groups to co-design guidelines on engagement with First Nations people on the REZs. Some respondents noted that in recent years the scope of consultation with First Nations communities on projects had broadened, and that some developers had begun engaging earlier (INT14B, INT20B). However, Norman et al. find that by mid-2023 there was still "limited and uneven engagement with Aboriginal communities on renewable projects in the Central-West Orana REZ and other declared REZs" [53]. Local Aboriginal Lands Councils in NSW reported that "they were not prepared or fully aware of the possibilities for engaging with renewable energy operators or projects on their land" [53].

4.2. Distributive equity

This section addresses the potential for renewable energy expansion under the Roadmap to produce socioeconomic benefits and impacts for local communities, encompassing broader economic consequences (4.2.1), benefit-sharing mechanisms (4.2.2), community ownership (4.2.3) and implications for First Nations people (4.2.4). Initial policy implementation is *coherent* if efforts to expand renewable energy are likely to enhance distributive equity, and *incoherent* if those efforts either exacerbate or fail to address distributive inequity.

4.2.1. Economic benefits and impacts within REZs

As noted in the introduction, one important function of REZs is to concentrate – both geographically and temporally – the benefits and impacts of renewable energy development within these zones. This poses both opportunities and challenges for coherent and equitable policy-making and implementation. When the Roadmap was launched, EnergyCo estimated that landholders hosting projects in the CWO REZ could receive up to A$430 million in periodic lease payments (INT15B, INT25N, [49]). Broader communities may reap economic benefits through employment, local procurement and increased economic activity associated with construction (INT11G, [54]) and benefit-sharing mechanisms (see section 4.2.2 below). When launching the Roadmap, the NSW Government highlighted its potential to generate up to A$1.5 billion in lease payments and substantial employment for regional communities, with an estimate of 6300 construction jobs and 2800 ongoing jobs by 2030, mostly in regional NSW in 2030 [11].

However, a range of adverse consequences could also accrue, including land use disruption (INT14B, INT26G, [49]) and visual/noise impacts (INT12N, INT21N, INT26G). Adverse economic impacts, while only briefly touched upon in the Roadmap document, were a key concern for many interviewees, including pressure on the availability and price of housing, healthcare and other services, limited availability of workers for other needs, traffic disruption, increased costs of road maintenance and waste disposal, and local biodiversity (INT10G, INT11G, INT20B, INT21N, INT22B, INT24N, INT26G; see also [48,49]). All of these impacts could accumulate due to the higher concentration of projects in REZs.

The NSW policy framework includes a range of measures to address impacts and benefits, including: state environmental and social impact assessment requirements; additional merit criteria for developers tendering for projects in REZs covering land use/community impacts and regional economic development; and bodies established under the EII Act such as the Renewable Energy Sector Board and the Electricity Infrastructure Jobs Advocate that advise on how to maximise economic, training and employment benefits from REZs. EnergyCo has also conducted analytical work on impacts and benefits and outlined opportunities to address them [55].

Some participants identified factors that could limit the ability of regional communities to benefit from renewable energy projects, including local skills shortages and low levels of unemployment (INT11G, INT22B, [49,54,56,57]). Accordingly, some respondents felt that the local employment and procurement targets set out in NSW guidelines were challenging (INT11G, INT22B) or even unrealistic (INT20B) given the limited local manufacturing capacity and

[9] Another recent example of co-design at the federal level is the development of the First Nations Clean Energy Strategy [50].

workforces. The NSW Renewable Energy Sector Board found in 2024 that renewable energy projects that had been awarded LTESAs were often struggling to meet local content requirements, particularly for the operations and maintenance phase [58].

While renewable energy projects in REZs could open up economic opportunities for regional areas, some groups that are under-represented in the renewable energy workforce or associated industries (including women, low-income groups and people with disabilities) may face obstacles in harnessing these opportunities (INT11G). LTESA tenders require that proponents commit to employing at least 15 % of their workforce from underrepresented groups,[10] but the Renewable Energy Sector Board found in 2024 that most project proponents had only made "minimal commitments" to this end [58]. Moreover, low-income groups may experience adverse impacts (e.g. higher rental prices or pressure on health services) disproportionately.

4.2.2. Benefit-sharing mechanisms

The NSW Government's Benefit-sharing Guideline for large-scale renewable energy projects recognises the importance of benefit-sharing mechanisms beyond individual landholder payments on the basis that "host communities, which bear the brunt of the changes, may not necessarily experience a proportionate level of benefits from the uptake of renewable energy" [59]. Benefit-sharing mechanisms for renewable energy schemes in NSW involve a mix of voluntary and mandatory components [49] and commonly support initiatives such as community infrastructure upgrades or sponsorship of local organisations (INT24N). The Roadmap has incentivised voluntary benefit-sharing arrangements through tender criteria as well as mandating that operators who obtain access rights pay access fees, a portion of which will be channelled towards community and employment purposes via REZ-wide schemes overseen by EnergyCo (INT10G).

Numerous respondents affirmed the importance of benefit-sharing mechanisms for building community support for renewable energy projects (INT12N, INT14B, INT17N, INT20B). However, some community members were concerned that monetary benefit-sharing mechanisms were either insufficient or inappropriate for offsetting some impacts (e.g. on landscapes or community cohesion), and there was a degree of suspicion that developers are "buying off the community to continue doing harm" (INT24N) or that payments largely serve to placate urban electricity consumers (INT31N).

REZ-wide benefit-sharing schemes offer the possibility of delivering more substantial benefits to communities (INT12N, INT24N). Ahead of the commencement of schemes funded by access fees, the NSW Government announced in 2023 an A$128 million Community Employment and Benefit Program in the CWO REZ, and in early 2025 announced the first A$60 million in grants under the program [60]. The largest single grants were allocated to local councils, notably to establish a training facility for the renewable energy workforce and housing for key workers in larger population centres, but a substantial amount of this funding also went towards upgrading infrastructure in smaller towns close to REZ projects. A much larger number of small grants was distributed among a variety of community organisations across the REZ. Of these, a small proportion of grants was targeted towards First Nations groups and culturally and linguistically diverse communities, and a larger proportion towards children and young people. The Program also includes a dedicated A$10 million First Nations Fund. While the sums allocated so far are fairly modest, they will increase over time as project-specific and REZ-wide schemes ramp up. Overall, the allocation of funds appears to reflect (at least in part) a mix of criteria for distribution identified by NSW policies and community members in the CWO REZ, including prioritising those most affected by REZ projects while also promoting broader-based regional development (see section 5.4.2).

4.2.3. Community ownership and co-investment

Community-owned renewable projects aim not only to augment renewable energy capacity but also to empower communities and build social licence for the energy transition (INT14B; see also FG02R, [48,61]). While community-owned renewable projects are prominent in some countries such as Germany, Denmark and Scotland, to date there are relatively few in Australia [49], although the sector has experienced significant growth over the past decade [62].

A respondent commented that the Roadmap neither blocks nor incentivises community ownership/investment models in the REZs (INT12N). In practice, however, several factors tend to reinforce the emphasis on large-scale projects run by private developers, including: limited technical capacity among communities and developers (INT17N); high financial risk for small projects particularly during their early stages (INT17N); and regulatory barriers to people directly using the energy from community energy projects (FG02R). As of mid-2025, no community-owned energy project groups were recorded as being active in the CWO REZ [63], but at least one project developer (Squadron Energy) has launched a program to enable community co-investment in its CWO REZ projects [64].

4.2.4. Benefits and impacts for First Nations people

Renewable energy projects could yield a range of benefits for First Nations people living in REZs (FG01R, FG02R, INT08B, INT18R, INT31N, [53]). Many of these benefits parallel those outlined in section 4.2.1 above, including employment, procurement and benefit-sharing mechanisms. Two further aspects include improving energy reliability, especially in more remote areas [53] (see also FG01R, FG02R, INT08B, INT18R, INT31N), and economic benefits associated with hosting projects on land held under land rights or Native Title legislation (INT18R).

Despite these opportunities, respondents also noted a range of challenges. Here we focus on two socio-economic aspects.[11] First, options for siting projects directly on Land Rights Act land in NSW were often limited (INT27N) (especially in comparison with jurisdictions such as Western Australia and the Northern Territory), as this land is often marginal, fragmented and unsuitable for renewable energy projects (e.g. due to conservation zoning; INT18R). Meanwhile, other parcels of land are subject to long-unresolved land claims (INT27N, INT28N). As of mid-2025 there were no large-scale Aboriginal-owned renewable energy projects in NSW [66]. Second, there are challenges associated with accessing employment opportunities. Aboriginal and Torres Strait Islander people remain underrepresented in the renewable energy workforce, comprising around 0.8 % of the total (despite making up 3.3 % of Australia's overall population: [67]). The Roadmap includes a range of mechanisms to encourage First Nations employment and procurement in REZ projects, including criteria in tenders for LTESAs. However, most projects have fallen short of these targets [58,66], indicating that this aspect of distributive equity has been difficult to fulfil in practice.

5. Drivers of policy coherence and incoherence

5.1. Section overview

In this section we investigate a range of factors that could explain the evidence of policy coherence and incoherence presented in the previous section, categorised across three dimensions: ideas, institutions and interests. For each dimension we select two drivers that emerged as most prominent in interviews and documentary analysis. Appendix A3 summarises the full range of drivers identified in the case study. Further analysis of each driver (including broader systemic drivers such as

[10] These include women, youth, long-term unemployed and people with characteristics defined under the Anti-Discrimination Act 1977 (NSW).

[11] Projects could have a range of adverse impacts on Aboriginal and Torres Strait Islander cultural heritage (INT15B, INT27N) but detailed discussion of this point is beyond the scope of this article (for further analysis see [65]).

neoliberalism and colonial legacies) can be found in [43] (see also [68–70]).

5.2. Ideas

5.2.1. Political ideologies: polarization and common ground

Ideological polarisation can drive incoherence by prioritising a narrow range of policy objectives while sidelining others [43]. Conversely, cross-party deliberation can inject greater ideological diversity into policy formulation, albeit at the potential cost of greater institutional complexity (see section 5.3.1 below) or a mismatch between the range of objectives espoused and the resources available to achieve them.

At the federal level, and to a lesser extent at the state level, Australia's two main political parties (the Liberal Party of Australia and the Australian Labor Party) hold divergent views on the energy transition and its social implications. Typically, governments led by the Liberal Party, often in coalition with the more climate-sceptical National Party, have been more reluctant to act on climate change than those led by the Australian Labor Party [68,71,72]. However, the state Coalition government which introduced the NSW Roadmap held a more proactive stance on climate change than the federal Coalition. This may be attributable to the demographics of NSW, which are predominantly urban, and the fact that fossil fuel extraction makes up a much smaller share of the state's economy than some other jurisdictions [73].

The NSW Coalition government sought cross-party support for the EII Act partly out of necessity (as it held a minority in the upper house of Parliament) but also to reduce the risk that the Act would be repealed under a future change of government [74]. In the resulting cross-party negotiations, Labor and the Greens proposed amendments that resulted respectively in the expansion of REZs to two key industrial regions and the adoption of guidelines on engagement with First Nations people ([75], INT18R). The negotiations promoted coherence by ensuring that a wider range of equity objectives were integrated into the Act. Following a change of government in 2023, the incoming Labor government reaffirmed its support for the Act but initiated a policy review that aimed to improve Roadmap implementation through increased resourcing and higher prioritisation on the government's agenda [38]. Another review held in early 2025 found that stakeholder support for the objectives of the Act remained strong and that concerns mainly revolved around its implementation in practice [42].

5.2.2. Conflicting ideas of scale

One common concern voiced by participants was that the scale of renewable energy infrastructure envisaged under the Roadmap is not well suited to the types of projects that would be most beneficial to First Nations communities (INT18R; see also FG01R, INT08B, INT09B) or regional communities generally (INT12N, INT17N):

> I think that there's a general unawareness about the potential for community scale, community owned, community led energy solutions. […] the state governments […] say, well, we've got these five big coal powered generators, we need to replace them with big business renewables. That's just how the mentality goes. It's not, what about we have lots of smaller ones, which would actually make our grid more resilient rather than just a few large-scale ones owned here and there. (INT17N)

Similarly, Proudlove et al. [76] argue that the scale of the National Electricity Market poses a significant barrier to the uptake of community-owned energy infrastructure. Numerous commentators argue that the urgency of the transition instead requires an "all of the above" approach in the sense that renewable energy infrastructure needs to expand at all scales (see e.g. [38]). However, privileging some scales over others can come at a cost for distributive equity.

5.3. Institutions

5.3.1. Institutional capacity, complexity and inertia

Institutional capacity across both government agencies and non-government organisations posed a major constraint on both the speed of the transition and the ability of those organisations to advance equity. When the Roadmap was launched, the NSW Government initially lacked the institutional capacity to deliver it, in part due to the legacy of privatisation, the shift of institutional responsibilities associated with the National Electricity Market, and limited recent experience with construction of transmission infrastructure [38]. EnergyCo's capacity to engage meaningfully with communities was particularly limited during the early phase of the Roadmap, although it improved to some degree as EnergyCo's staffing profile built up (INT12N, INT26G). Institutional capacity was also a concern for other organisations, including local councils (INT26G), project developers (INT14B, INT20B), Aboriginal Land Councils (INT18R, INT27N, INT28N) and community organisations (INT23N, INT24N).

Institutional complexity emerged as a recurring obstacle both to the government's ability to coordinate Roadmap implementation and the ability of people to participate in decision-making (INT13N, INT22B, [38]; see also [42] and section 4.1.3). The complexity of the Roadmap framework arises not only from overlaying the REZ model upon an already complex thicket of federal/state regulation [36] but also from cross-party compromises struck during negotiations on the Act that established additional bodies for the benefit of workers and regional development (see section 5.2.1).

5.3.2. Institutional coordination

Recall from section 2.1 that horizontal coherence (across policy objectives at a single level of governance) may interact with vertical coherence (across levels of government). A key insight from the NSW case study and related research is that a government's ability to achieve coherence between energy and equity objectives may be enhanced or impeded by the extent to which it can coordinate effectively (a) among different agencies at the same level; and (b) with other levels of government.

Horizontally, challenges to coherence arise in coordinating across departments at each level of government. Longstanding silos between climate- and energy-related institutions (and within the electricity sector, e.g. between regulating new transmission and generation) in Australia have constrained opportunities to resolve tensions or identify synergies across policy objectives [36,72]). While the EII Act included some coordination measures from the start (such as the Electricity Infrastructure Jobs Advocate and the Renewable Energy Sector Board) it soon became apparent that a more extensive range of measures to coordinate across departments was required, leading to the establishment of bodies such as a Roadmap Intergovernmental Steering Committee [38] as well as operational agreements between agencies and cross-agency reviews. Nevertheless, O'Reilly and Nemes [38] found that in mid-2023 a "siloed approach" to coordinating the transition continued to prevail. Different parts of the NSW Government have given conflicting signals at times on issues such as planning approval and the suitability of land for wind farm development (e.g. INT20B, [77]).

Vertically, coordination across government levels can enhance or impede coherence. Despite the intergovernmental coordination arrangements established under the National Electricity Market, coherence across state and federal levels has often been difficult to achieve, particularly when different political parties are in power at each level (see section 5.2.1 above). Prospects for coordination became more favourable with the change of Federal Government in 2022 (INT13N). An example of where state and federal coordination can strengthen synergies between renewable energy and equity is the National Energy Transformation Partnership launched in 2022 by Commonwealth, State and Territory Energy Ministers [78]. One of two priority actions in the Partnership's inaugural document was to "Co-design a new First Nations

Clean Energy Strategy, to ensure First Nations people in Australia shape and drive the clean energy transformation." The Strategy, published in 2024, complements state-level initiatives such as the First Nations guidelines mentioned in section 4.1.5. Despite these positive developments, there have been ongoing challenges in coordinating the financial contributions of the Federal Government and state-based schemes towards the construction of REZs, as well as varying regulatory requirements for community engagement across state and federal legislation [79].

5.4. Interests

5.4.1. Energy industry interests

The coherence of energy policies in Australia has long been hampered by the fossil fuel industry, which has persistently (and often misleadingly) sought to portray renewable energy measures as undermining economic growth, employment and regional prosperity [72,80]. Here we focus instead on how the interests of the renewable energy industry may influence the ability of the REZ model to incorporate equity objectives.

Solar and wind project developers have both commercial and regulatory incentives to engage with communities and deliver benefits (INT12N, INT14B, INT20B, [81]), some of which are heightened by the REZ model. Demonstrating community engagement and support along with a commitment to benefit-sharing is simultaneously: a criterion for obtaining rights to operate in REZs as well as planning approval (INT14B); an important element in maintaining social licence (INT20B); and an opportunity for developers to stand out in a highly competitive sector (INT12N; INT20B). At the same time, interviewees pointed out that there are limits to companies' willingness to invest in community engagement and benefit-sharing, not least due to their foundational profit motive as well as government guidelines which set maximum amounts for benefit-sharing [59].

Competition between developers can be a double-edged sword when it comes to benefit-sharing. A representative of a First Nations organisation pointed out that this competition can be leveraged to improve outcomes for communities (INT27N). However, competition can impede coordinated and higher-impact allocation of benefit-sharing resources, as some developers prefer to ensure that the benefits they provide are individually branded (INT15B).

5.4.2. Interests within and across communities

The extent of alignment between communities' interests in advancing the energy transition and social equity may significantly affect the coherence of REZ implementation. Community members have concurrent interests in securing benefits from renewable energy projects while avoiding their adverse impacts (INT12N, INT25N). One survey conducted in 2024 found that around 69 % of people living in REZs believed that "local communities stand to gain when their local area hosts renewable energy projects if things are done right" [82], while another held in early 2025 found that people living in existing or potential REZs were less confident than other Australians that they will benefit from the energy transition [83].

A coherent approach is made more challenging by the fact that community members' interests may clash. Sometimes this may be a product of landowners competing to host a project or disagreeing with neighbours who wish to do so (INT26G, INT25N). Interests may also clash over how benefits should be distributed. The NSW Government's Benefit-Sharing Guideline identifies degree of impact and the size of affected populations as primary criteria for distributing benefits [59]. However, these two criteria may come into tension, as reflected in concerns raised by community members about larger but less affected population centres receiving substantial shares of funding. A survey of 147 respondents about the CWO REZ's Community and Employment Benefits Program found that over 83 % believed that some communities should be prioritised over others in benefit-sharing, primarily for equity reasons, while a majority thought that degree of impact (which primarily relates to smaller localities) should be the most important criterion [84]. However, respondents cited a number of different principles for distribution. A diversity of views may not in itself produce incoherence, but the procedural avenues chosen to inform decision-making (ranging from surveys of the kind used for this program to more co-designed approaches such as participatory budgeting [85]) may influence public perceptions of whether the resulting distribution is seen as fair.

6. Discussion and policy implications

The results presented in section 4 reveal elements of coherence and incoherence for both procedural and distributive equity, as summarised in Fig. 1 below. Some aspects of (in)coherence reflect findings from previous research on individual renewable energy projects, e.g. variations in the quantity and quality of consultation may influence the extent to which procedural equity is enhanced or undermined by the projects (e.g. [86,87]). The analysis also reinforces findings that procedural and distributive equity may interact to produce more or less coherent outcomes: for example, the extent of community participation in decisions about benefit-sharing and impact management may influence the extent to which those decisions align with public perceptions of distributive fairness [13,70,88].

Equally, the analysis also highlights the acuteness of the challenge of coherence in REZs in three respects. First, the larger number of proposed projects in a REZ increases both the amount of information that affected communities need to process and act upon, and the extent of institutional capacity needed to ensure meaningful and coordinated community engagement. Second, institutional complexity resulting from overlaying the REZ model upon existing institutions can inadvertently pose obstacles to procedural equity by rendering the system harder to navigate. Third, the intensification of benefits and impacts in REZs opens up new opportunities but also increases risks of inequitable distribution. The intensity of these challenges is underscored by the fact that while benefit-sharing mechanisms have begun to flow to communities in the CWO REZ, overall progress on the Roadmap remains behind schedule, partly due to growing community opposition [5], which may in turn have been affected by prior public engagement processes.

The analysis in section 5 indicates that coherence and incoherence can be driven by a mix of ideas, institutions and interests, and that these dimensions may interact. For example, institutional factors such as regulatory measures may change the way in which industries pursue their interests, while a mix of ideas, institutions and interests can converge to privilege large-scale, corporate-owned projects over community-owned initiatives (see also [2]).

While it is difficult to draw precise conclusions about the dominance of one type of driver over others, Fig. 1 shows that some drivers affected multiple aspects of (in)coherence, notably institutional capacity and industry interests. Some respondents argued that interests ultimately mattered more than ideas or institutions (e.g. INT14B). Even if this view is plausible (which remains a matter for debate), this does not mean that interests remain static or that ideas and institutions are irrelevant. While fossil fuel interests continue to obstruct a swift and equitable energy transition [89], the lower cost of renewables is shifting the configuration of industry and community interests, and our analysis reveals some potential to harness renewable energy interests not only to accelerate the transition but also to make it more equitable (see section 5.4.1).

Interviews and documentary analysis revealed a range of policy options that could enhance coherence. While space precludes a comprehensive account of these options, a selection of key recommendations for current and future REZs is summarised in Table 1 below.[12] The recommendations are formulated to be applicable both to Australia

[12] A more comprehensive set of policy recommendations can be found in [43].

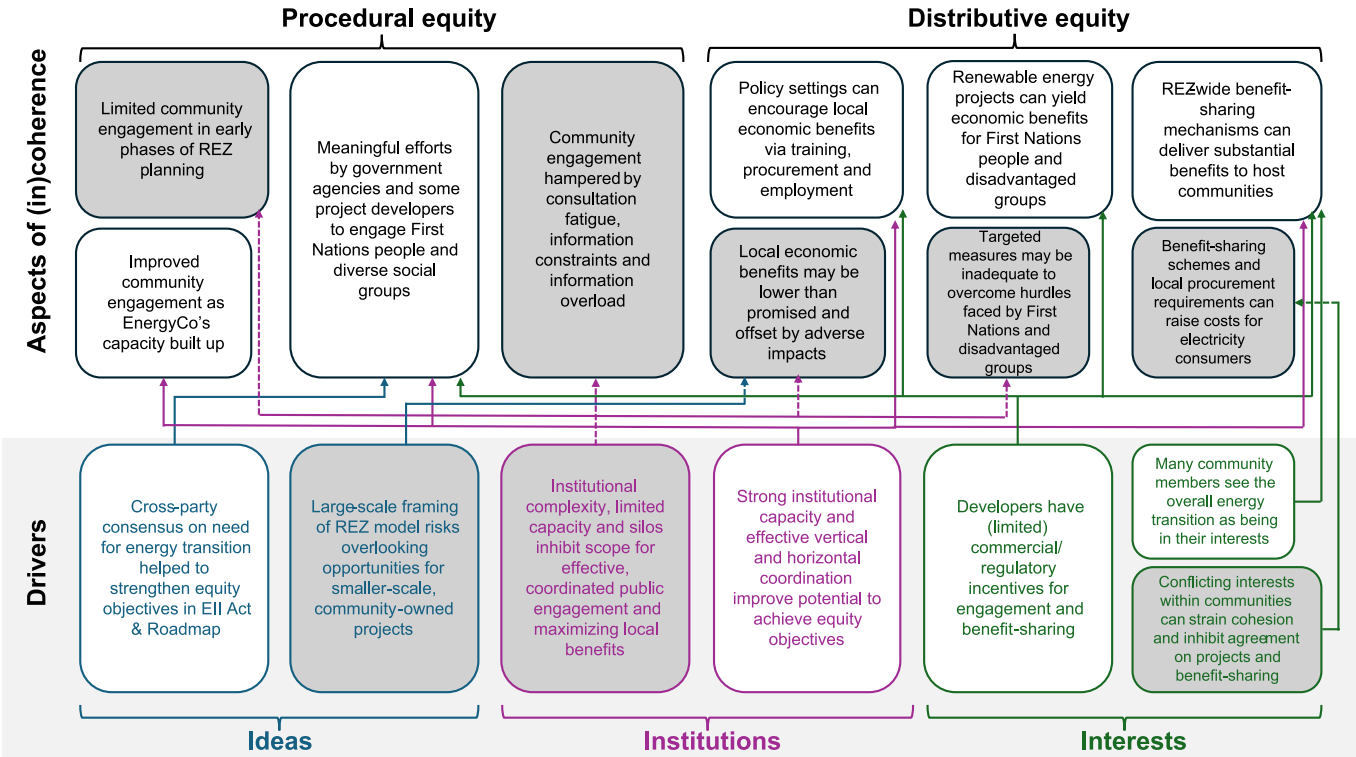

Fig. 1. Aspects of (in)coherence and their associated drivers in the NSW Roadmap. Aspects/drivers of coherence have a white background (connected with a solid arrow) while aspects/drivers of incoherence have a grey background (connected with a dashed arrow).

and internationally, spanning most areas of procedural and distributive inequity identified in section 4, while also targeting one or more key drivers identified in section 5. A number of these recommendations are already being pursued to varying degrees in NSW (e.g. REZ-wide benefit-sharing schemes), but the recommendations also highlight ongoing areas for improvement (e.g. benefits for First Nations groups and local employment). Future implementation should be informed by (and can in turn inform) emerging knowledge on best practices in areas such as community engagement, Indigenous participation [19,65] and benefit-sharing [88].

7. Conclusion

Renewable energy zones offer opportunities to coordinate and accelerate the energy transition while advancing the wellbeing of regional communities and broader populations. However, our analysis has shown that REZs also pose significant challenges for policy coherence, particularly due to the concentration of socioeconomic impacts within these zones and the institutional complexity that can arise when REZs are overlaid upon existing policy frameworks. Thus, the promise of "regional equity zones" remains a work in progress. Our aim in this article has not been to assess the overall merits of REZs compared to more decentralised models, but rather to show how a policy coherence perspective can shed light on their advantages and limitations, with a view to informing future policy discussions in this area.

Further research would be valuable on drivers of coherence and incoherence relating to other elements of the energy transition (e.g. electricity transmission), particular social groups (e.g. First Nations, women and low-income households), and other policy objectives (e.g. biodiversity) [90]. Additional research on REZs in Australia and beyond should also track if and how (in)coherence varies over time from the earliest stages of REZ development through to the construction and post-construction phases, particularly as some promised socio-economic benefits may or may not materialise, while unexpected synergies or trade-offs could emerge. This will require both qualitative (e.g. attitudinal) research of the kind presented in this article as well as quantitative analysis (e.g. of local impacts on training, employment, procurement and benefit-sharing).

While some features of coherence and incoherence in the present case study may be found in other jurisdictions in Australia and overseas, the present study suggests that the dynamics of policy (in)coherence may be highly contextual, particularly when it comes to variable factors such as government ideologies, institutional capacity and community values and preferences. Comparative research could yield valuable knowledge on what drives (in)coherence and on ways of achieving greater coherence in energy policy.

CRediT authorship contribution statement

Jonathan Pickering: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Pierrick Chalaye:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization.

Funding sources

This research was funded by the Swedish Research Council for Sustainable Development (Formas) project "National implementation of global goals: Coherence between national climate plans and global sustainability goals" (Grant number 2018-01706), with supplementary funding from the research support scheme of the Faculty of Business, Government and Law at the University of Canberra.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Table 1
Recommendations for policy-makers and project developers to advance a coherent and equitable energy transition in REZs (with cross-references to related sections of the article).

Recommendation	Types of equity advanced	Drivers of (in) coherence addressed
1. Build cross-party support for equitable approaches to REZ construction that can secure widespread public buy-in and future-proof reforms	• Procedural (4.1.1–4.1.2) • Distributive (4.2.1–4.2.4)	• Ideas (5.2.1)
2. High-level ministerial and bureaucratic leadership and effective inter-agency coordination to integrate equity for REZ communities and other participants in the electricity system	• Procedural (4.1.1–4.1.5) • Distributive (4.2.1–4.2.4)	• Institutions (5.3.1, 5.3.2)
3. Policies and practices that incentivise government agencies and project developers to implement early, authentic, inclusive, adequately resourced, sustained and coordinated engagement with First Nations groups and communities	• Procedural (4.1.1–4.1.5)	• Institutions (5.3.1, 5.3.2) • Interests (5.4.1, 5.4.2)
4. Project design and coordinated policy initiatives that (a) minimise adverse impacts, with special attention to cumulative impacts produced by the REZ model; and (b) maximise local economic benefits (e.g. through training, employment, procurement or improving energy affordability), particularly for heavily impacted areas and for First Nations and underrepresented groups	• Distributive (4.2.1–4.2.4)	• Institutions (5.3.1, 5.3.2) • Interests (5.4.2)
5. Stronger regulatory and financial incentives for communities and First Nations groups to own and invest in REZ projects	• Procedural (4.1.2, 4.1.3, 4.1.5) • Distributive (4.2.3, 4.2.4)	• Ideas (5.2.2) • Interests (5.4.1, 5.4.2)
6. Inclusive design of both REZ-wide and project-specific benefit-sharing schemes that deliver meaningful, sustained benefits, particularly for residents heavily impacted by projects, First Nations people and disadvantaged groups	• Procedural (4.1.3, 4.1.4, 4.1.5) • Distributive (4.2.2, 4.2.4)	• Institutions (5.3.2) • Interests (5.4.1, 5.4.2)
7. Rigorous and timely monitoring, evaluation, learning shared with affected communities which tracks cumulative impacts and benefits as well as synergies and trade-offs, and which holds governments and developers accountable for their performance against energy and equity objectives	• Procedural (4.1.1–4.1.5) • Distributive (4.2.1–4.2.4)	• Institutions (5.3.1, 5.3.2)

Jonathan Pickering reports financial support was provided by Swedish Research Council for Sustainable Development (Formas). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grateful for extensive and constructive support and advice from the project coordination team at the Stockholm Environment Institute, in particular Adis Dzebo, Zoha Shawoo and Katherine Browne. We presented earlier drafts of this work at the Centre for Governance and Public Policy at Griffith University, the Earth System Governance conference at Radboud University, the Australian Political Studies Association conference, and the Canberra School of Politics, Economics and Society and the Centre for Environmental Governance at the University of Canberra. We are very grateful to audiences at these events for their constructive comments and questions. For advice, feedback and support throughout the project we would also like to thank our colleagues John Dryzek, Liam Mobbs (who worked with the team as a research intern on aspects of the New South Wales policy framework), Jo Mummery, Elise Remling and Darren Sinclair at the University of Canberra, and Suhasini Gunatillaka, Susan Park and David Schlosberg at the Sydney Environment Institute. We appreciate the feedback of two anonymous reviewers for the journal which prompted significant improvements to the manuscript. Finally, many thanks to our interview and focus group participants for sharing their experiences and insights.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.erss.2025.104415.

Data availability

The data that has been used is confidential.

References

[1] P. Joshi, J. Desai, Transmission planning with renewable energy zones (REZ): overview, select case studies, and key enablers, Clean, Energy Ministerial (CEM) Senior Officials Meeting, https://research-hub.nrel.gov/en/publications/transmission-planning-with-renewable-energy-zones-rez-overview-se, 2023.
[2] L. Baker, A. Hook, B.K. Sovacool, Power struggles: governing renewable electricity in a time of technological disruption, Geoforum 118 (2021) 93–105, https://doi.org/10.1016/j.geoforum.2020.12.006.
[3] Australian Energy Infrastructure Commissioner, Community Engagement Review Report, Department of Climate Change, Energy, the Environment and Water, Canberra, https://www.dcceew.gov.au/sites/default/files/documents/community-engagement-review-report-minister-climate-change-energy.pdf, 2023.
[4] D. Cass, L. Connor, R. Heikkinen, R. Pearse, Renewables & Rural Australia: A Study of Community Experiences in Renewable Energy Zones in NSW and the Case for More Equity and Coordination of the Clean Energy Transformation, Sydney Environment Institute, The Australia Institute, 2022. https://australiainstitute.org.au/wp-content/uploads/2022/06/P1176-Renewables-and-Rural-Australia-appendix-Dec-22.pdf, 2023.
[5] Net Zero Commission, 2024 Annual Report. https://www.netzerocommission.nsw.gov.au/sites/default/files/2024-11/Net_Zero_Commission_2024_Annual_Report.pdf, 2024.
[6] L. Susskind, J. Chun, A. Gant, C. Hodgkins, J. Cohen, S. Lohmar, Sources of opposition to renewable energy projects in the United States, Energy Policy 165 (2022) 112922, https://doi.org/10.1016/j.enpol.2022.112922.
[7] G. Sanchez Nieminen, E. Laitinen, Understanding local opposition to renewable energy projects in the Nordic countries: a systematic literature review, Energy Res. Soc. Sci. 122 (2025) 103995, https://doi.org/10.1016/j.erss.2025.103995.
[8] A. Lenschow, P. Bocquillon, L. Carafa, Understanding coherence between policy spheres, Environ. Policy Gov. 28 (5) (2018) 323–328, https://doi.org/10.1002/eet.1818.
[9] M. Nilsson, N. Weitz, Governing trade-offs and building coherence in policy-making for the 2030 agenda, Politics & Governance 7 (4) (2019) 10, https://doi.org/10.17645/pag.v7i4.2229.
[10] Z. Shawoo, A. Maltais, A. Dzebo, J. Pickering, Political drivers of policy coherence for sustainable development: an analytical framework, Environ. Policy Gov. 33 (4) (2023) 339–350, https://doi.org/10.1002/eet.2039.
[11] New South Wales Government, Electricity Infrastructure Roadmap. https://www.energy.nsw.gov.au/nsw-plans-and-progress/major-state-projects/electricity-infrastructure-roadmap, 2020.
[12] B.K. Sovacool, M. Burke, L. Baker, C.K. Kotikalapudi, H. Wlokas, New frontiers and conceptual frameworks for energy justice, Energy Policy 105 (2017) 677–691, https://doi.org/10.1016/j.enpol.2017.03.005.
[13] P. Devine-Wright, F. Sherry-Brennan, Where do you draw the line? Legitimacy and fairness in constructing community benefit fund boundaries for energy infrastructure projects, Energy Res. Soc. Sci. 54 (2019) 166–175, https://doi.org/10.1016/j.erss.2019.04.002.
[14] K. Gürtler, Justice in energy transformations as a spatial phenomenon: a framework for analyzing multi-dimensional justice claims, Energy Res. Soc. Sci. 105 (2023) 103277, https://doi.org/10.1016/j.erss.2023.103277.
[15] World Energy Council, World Energy trilemma 2024: evolving with resilience and justice, World Energy Council (2024) https://www.worldenergy.org/assets/downloads/World_Energy_Trilemma_2024_Full_Report.pdf?v=1733924639.

[16] J. Rinkinen, E. Shove, The energy trilemma, in: J. Rinkinen, E. Shove, J. Torriti (Eds.), Energy Fables: Challenging Ideas in the Energy Sector, Routledge, 2019, pp. 91–102.
[17] OECD, Policy Coherence for Sustainable Development 2019: Empowering People and Ensuring Inclusiveness and Equality, 2019, https://doi.org/10.1787/a90f851f-en.
[18] S.E. Sharma, R.R. Das, A. Janzwood, N. Joshi, J.L. MacArthur, G. Savvidou, Equity, diversity and inclusion promises, exclusive practices? How to move towards effective and just energy transitions, Energy Res. Soc. Sci. 120 (2025) 103935, https://doi.org/10.1016/j.erss.2025.103935.
[19] C.E. Hoicka, How do we practice equity, diversity and inclusion in sustainable energy research? Advice for modern researchers, Energy Res. Soc. Sci. 97 (2023) 102964, https://doi.org/10.1016/j.erss.2023.102964.
[20] S. Droubi, R.J. Heffron, D. McCauley, A critical review of energy democracy: a failure to deliver justice? Energy Res. Soc. Sci. 86 (2022) 102444 https://doi.org/10.1016/j.erss.2021.102444.
[21] E. Baker, S. Carley, S. Castellanos, D. Nock, J.F. Bozeman III, D. Konisky, C. G. Monyei, M. Shah, B. Sovacool, Metrics for decision-making in energy justice, Annu. Rev. Environ. Resour. 48 (1) (2023) 737–760, https://doi.org/10.1146/annurev-environ-112621-063400.
[22] P. Graham, J. Hayward, J. Foster, in: CSIRO (Ed.), GenCost 2023-24: final report, 2024. https://www.csiro.au/-/media/Energy/GenCost/GenCost2023-24Final_20240522.pdf.
[23] A. Yunita, F. Biermann, R.E. Kim, M.J. Vijge, The (anti-) politics of policy coherence for sustainable development in the Netherlands: logic, method, effects, Geoforum 128 (2022) 92–102, https://doi.org/10.1016/j.geoforum.2021.12.002.
[24] C. Parsons, Showing ideas as causes: the origins of the European Union, Int. Organ. 56 (1) (2002) 47–84. https://www.jstor.org/stable/3078670.
[25] O.R. Young, L.A. King, H. Schroeder, Institutions and Environmental Change: Principal Findings, Applications, and Research Frontiers, MIT Press, Cambridge, MA, 2008.
[26] R.A. Huber, M. Tomas, S. Kacper, S. Ćetković, Is populism a challenge to European energy and climate policy? Empirical evidence across varieties of populism, J. Eur. Publ. Policy 28 (7) (2021) 998–1017, https://doi.org/10.1080/13501763.2021.1918214.
[27] L. Hughes, J. Urpelainen, Interests, institutions, and climate policy: explaining the choice of policy instruments for the energy sector, Environ. Sci. Pol. 54 (2015) 52–63, https://doi.org/10.1016/j.envsci.2015.06.014.
[28] C. Strambo, M. Nilsson, A. Månsson, Coherent or inconsistent? Assessing energy security and climate policy interaction within the European Union, Energy Res. Soc. Sci. 8 (2015) 1–12, https://doi.org/10.1016/j.erss.2015.04.004.
[29] A. Finkel, K. Moses, C. Munro, T. Effeney, M. O'Kane, Independent Review into the future security of the National Electricity Market: blueprint for the future, Commonwealth of Australia (2017). https://www.dcceew.gov.au/energy/publications/independent-review-future-security-national-electricity-market-blueprint-future.
[30] R. Wilkins, E. Vera-Toscano, F. Botha, M. Wooden, T.-A. Trinh, The Household, Income and Labour Dynamics in Australia Survey: Selected Findings from Waves 1 to 20, Applied Economic & Social Research, University of Melbourne, Melbourne Institute, 2022 https://melbourneinstitute.unimelb.edu.au/_data/assets/pdf_file/0011/4382057/HILDA_Statistical_Report_2022.pdf.
[31] R. Tanton, L. Dare, R. Miranti, Y. Vidyattama, A. Yule, M. McCabe, Dropping off the edge 2021: persistent and multilayered disadvantage in Australia, Jesuit Social Services, 2021. https://www.dote.org.au/s/DOTE_Report-_Final.pdf.
[32] L. Chancel, T. Piketty, E. Saez, G. Zucman, World Inequality Report 2022, Harvard University Press, 2022.
[33] Australian Government, National agreement on closing the gap. https://www.closingthegap.gov.au/national-agreement/national-agreement-closing-the-gap, 2020.
[34] T. Longden, S. Quilty, B. Riley, L.V. White, M. Klerck, V.N. Davis, N.F. Jupurrurla, Temperature extremes exacerbate energy insecurity for indigenous communities in remote Australia, Nat. Energy 7 (1) (2022) 11–12, https://doi.org/10.1038/s41560-021-00968-6.
[35] DCCEEW, Powering Australia, Department of Climate Change, Energy, the Environment and Water, https://www.energy.gov.au/government-priorities/australias-energy-strategies-and-frameworks/powering-australia, 2023.
[36] A. Kallies, The Australian energy transition as a federalism challenge: (un) cooperative Energy federalism? Transnat. Environ. Law 10 (2) (2021) 211–235, https://doi.org/10.1017/S204710252000045X.
[37] Australian Bureau of Statistics, National, state and territory population, Retrieved 24 July 2025 from, https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/latest-release, 2025.
[38] C. O'Reilly, V. Nemes, Electricity supply and reliability check up, Marsden Jacob (2023). https://www.energy.nsw.gov.au/sites/default/files/2023-09/NSW_Electricity_Supply_and_Reliability_CheckUp_Marsden_Jacob_Report_2023.pdf.
[39] Australian Government, States and territories. Department of Climate Change, Energy, the Environment and Water. https://www.energy.gov.au/data/states-and-territories, 2023.
[40] Clean Energy Council, Clean Energy Australia: Report 2023. https://assets.cleanenergycouncil.org.au/documents/Clean-Energy-Australia-Report-2023.pdf, 2023.
[41] NSW Government, NSW electricity infrastructure roadmap: building an Energy superpower: detailed report. https://www.energy.nsw.gov.au/sites/default/files/2022-08/NSW%20Electricity%20Infrastructure%20Roadmap%20-%20Detailed%20Report.pdf, 2020.
[42] R. Owens, D. Swift, C. Rozyn, G. Swier, J. Hunt, NSW transmission planning review: Interim Report. https://www.energy.nsw.gov.au/sites/default/files/2025-06/nsw-transmission-planning-review-interim-report.pdf, 2025.
[43] J. Pickering, P. Chalaye, Towards a coherent energy transition: expanding renewable energy and reducing inequalities in Australia, Centre for Environmental Governance, Centre for Deliberative Democracy and Global Governance, University of Canberra, https://www.centreforenvironmentalgovernance.com/research/research-report%3A-towards-a-coherent-energy-transition%3A-expanding-renewable-energy-and-reducing-inequalities-in-australia, 2023.
[44] J. Penneman, A. Buchmayr, L. Van Ootegem, E. Verhofstadt, The evolution of the pre- and post-construction public opinions toward offshore wind energy on the Belgian coast, J. Environ. Plan. Manag. 66 (12) (2023) 2536–2555, https://doi.org/10.1080/09640568.2022.2079078.
[45] S. Fast, W. Mabee, Place-making and trust-building: the influence of policy on host community responses to wind farms, Energy Policy 81 (2015) 27–37, https://doi.org/10.1016/j.enpol.2015.02.008.
[46] NSW Government, NSW government submission on AEMO'S Integrated System Plan. https://aemo.com.au/-/media/files/electricity/nem/planning_and_forecasting/isp/2018/round-2-submissions/nsw-government—submission-on-integrated-system-plan.pdf?la=en, 2018.
[47] EnergyCo, Central-West Orana Renewable Energy Zone Transmission project: Submissions Report. https://www.energyco.nsw.gov.au/sites/default/files/2024-12/Submissions%20Report_0.pdf, 2024.
[48] EnergyCo, Community feedback report: Coordinating impacts and benefits in the Central-West Orana Renewable Energy Zone. https://www.energyco.nsw.gov.au/sites/default/files/2023-06/cwo-rez-report-community-feedback.pdf, 2023.
[49] NSW Agriculture Commissioner, Renewable energy generation and agriculture in NSW'S rural landscape and economy – growth sectors on a complementary path. https://www.dpi.nsw.gov.au/_data/assets/pdf_file/0005/1449860/210395fd12ea058abf3b424f4370204d64e105bb.pdf, 2022.
[50] Australian Government, First Nations Clean Energy Strategy. https://www.energy.gov.au/sites/default/files/2024-12/First%20Nations%20Clean%20Energy%20Strategy.pdf, 2024.
[51] RE-Alliance. (n.d.). How Hay approached renewable energy on their terms. Retrieved 7 April from https://www.re-alliance.org.au/hay_renewables_on_their_terms.
[52] M. Cellini, S. Loos, C. Mirenda, L. Pisacane, C. Striebing, S. Tagliacozzo, Exploring the nexus of gender and energy transitions: a systematic literature review, Energy Res. Soc. Sci. 119 (2025) 103887, https://doi.org/10.1016/j.erss.2024.103887.
[53] H. Norman, C. Briggs, T. Apolonio, Advancing Aboriginal interests in the New South Wales renewable energy transition, CAEPR Discussion Paper, Centre for Aboriginal Economic Policy Research, 2023, http://hdl.handle.net/1885/287128.
[54] C. Briggs, A. Atherton, J. Gill, R. Langdon, J. Rutovitz, K. Nagrath, Building a 'fair and fast' energy transition? Renewable energy employment, skill shortages and social licence in regional areas, Renew. Sustain. Energy Transit. 2 (2022) 100039, https://doi.org/10.1016/j.rset.2022.100039.
[55] EnergyCo, Central-West Orana Renewable Energy Zone: coordinating community impacts and benefits in the REZ. https://www.energyco.nsw.gov.au/sites/default/files/2023-03/cwo-rez-report-research-impacts-benefits.pdf, 2023.
[56] NSW Electricity Infrastructure Jobs Advocate, Electricity infrastructure jobs advocate's first report to the minister for Energy, NSW Treasury (2022). https://www.energy.nsw.gov.au/sites/default/files/2022-11/electricity-infrastructure-jobs-advocates-first-report-to-minister-for-energy-for-publication.pdf.
[57] A. Rainnie, D. Snell, Renewable energy and the promise of jobs, regional regeneration and first nations opportunities, Labour Ind. 33 (4) (2023) 403–421, https://doi.org/10.1080/10301763.2023.2289097.
[58] Renewable Energy Sector Board, Review of the Board's plan and recommendations to the Minister about implementation of the plan. https://www.energy.nsw.gov.au/nsw-plans-and-progress/major-state-projects/electricity-infrastructure-roadmap/roadmap-entities/sector-board#review-of-the-boards-plan, 2024.
[59] Department of Planning Housing and Infrastructure, Benefit-sharing guideline: guidance for large-scale renewable energy projects. https://www.planning.nsw.gov.au/sites/default/files/2024-11/benefit-sharing-guideline.pdf, 2024.
[60] P. Sharpe, T. Moriarty, $60 million in community benefits start flowing to the Central-West Orana Renewable Energy Zone (11 April 2025). https://www.energyco.nsw.gov.au/news/60-million-community-benefits-start-flowing-central-west-orana-renewable-energy-zone, 2025.
[61] RE-Alliance, Community benefits handbook. https://d3n8a8pro7vhmx.cloudfront.net/vicwind/pages/2630/attachments/original/1630468903/RE-Alliance_Community_Benefits_Handbook_WEB_01v1_%281%29.pdf?1630468903, 2021.
[62] E. Buckley, K. Walters, J.P. Marshall, A. Ford, in: Community Power Agency (Ed.), Australian community energy collective impact assessment 2023, 2023. https://cpagency.org.au/wp-content/uploads/2023/12/Community-Energy-Collective-Impact-Report_2023.pdf.
[63] Community Power Agency, Community energy map. https://cpagency.org.au/resources/community-energy-map/, 2025.
[64] Squadron Energy, Squadron Energy Central West Orana Community Investment Fund. https://www.squadronenergy.com/news/domacom-launches-squadron-energy-central-west-orana-community-investment-fund, 22 March 2024.
[65] L. O'Neill, K. Thorburn, B. Riley, G. Maynard, E. Shirlow, J. Hunt, Renewable energy development on the Indigenous estate: free, prior and informed consent and best practice in agreement-making in Australia, Energy Res. Soc. Sci. 81 (2021) 102252, https://doi.org/10.1016/j.erss.2021.102252.
[66] H. Norman, C. Briggs, E. Langham, T. Apolonio, S. Miyake, S. Niklas, S. Teske, Local Aboriginal Land Council Powershift: Sharing the Benefits of the Energy Transition (Policy Insights Paper, Issue). Australian Public Policy Institute.

https://appi.org.au/wp-content/uploads/2025/06/APPI-Policy-Insights-Paper-Local-Aboriginal-Land-Council-Powershift.pdf, 2025.
[67] Clean Energy Council, Empowering everyone: diversity in the Australian clean energy sector. https://assets.cleanenergycouncil.org.au/documents/resources/reports/Empowering-Everyone-Diversity-in-the-Australian-Clean-Energy-Sector.pdf, 2021.
[68] R. Pearse, Pricing Carbon in Australia: Contestation, the State and Market Failure, Taylor & Francis, 2017.
[69] S. Chandrashekeran, Rent and reparation: how the law shapes Indigenous opportunities from large renewable energy projects, Local Environ. 26 (3) (2021) 379–396, https://doi.org/10.1080/13549839.2020.1861590.
[70] F. Rioux-Gobeil, A. Thomassin, A just energy transition for Indigenous peoples: from ideal deliberation to fairness in Canada and Australia, Energy Res. Soc. Sci. 114 (2024) 103593, https://doi.org/10.1016/j.erss.2024.103593.
[71] K. Crowley, Fighting the future: the politics of climate policy failure in Australia (2015–2020), WIREs Clim. Change 12 (5) (2021) e725, https://doi.org/10.1002/wcc.725.
[72] R. MacNeil, Swimming against the current: Australian climate institutions and the politics of polarisation, Environ. Polit. (2021) 1–22, https://doi.org/10.1080/09644016.2021.1905394.
[73] ABS, Australian National Accounts: State Accounts. https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-state-accounts/latest-release, 2022.
[74] P. Hannam, Clean and green: powering through 'watershed week' for energy policy, Syd. Morning Herald, https://www.smh.com.au/environment/climate-change/clean-and-green-powering-through-watershed-week-for-energy-policy-20201127-p56ikc.html, 2020, 28 November.
[75] P. Hannam, One nation's Latham pushes 249 amendments to stymie NSW energy bill, Sydney Morning Herald, https://www.smh.com.au/environment/climate-change/one-nation-s-latham-pushes-249-amendments-to-stymie-nsw-energy-bill-20201119-p56g7b.html, 2020, 20 November.
[76] R. Proudlove, S. Finch, S. Thomas, Factors influencing intention to invest in a community owned renewable energy initiative in Queensland, Australia, Energy Policy 140 (2020) 111441, https://doi.org/10.1016/j.enpol.2020.111441.
[77] P. Hannam, Windfarm development could be even harder under NSW plan, renewables sector fears, Guardian Australia, https://www.theguardian.com/australia-news/2023/nov/16/windfarm-development-could-be-even-harder-under-nsw-plan-renewables-sector-fears, 2023, 16 November.
[78] Commonwealth State and Territory Energy Ministers, National Energy Transformation Partnership. https://www.energy.gov.au/government-priorities/energy-and-climate-change-ministerial-council/priorities/national-energy-transformation-partnership, 2022.
[79] G. Parkinson, "Deal sealed" for Australia's first renewable energy zone, but wait goes on for news of access rights, RenewEconomy, https://reneweconomy.com.au/deal-sealed-for-australias-first-renewable-energy-zone-but-wait-goes-on-for-news-of-access-rights/amp/, 2025, 4 April.
[80] M. Hudson, 'A form of madness': Australian climate and energy policies 2009–2018, Environ. Politics 28 (3) (2019) 583–589, https://doi.org/10.1080/09644016.2019.1573522.
[81] R.M. Colvin, G.B. Witt, J. Lacey, K. Witt, The community cost of consultation: Characterising the qualitative social impacts of a wind energy development that failed to proceed in Tasmania, Australia, Environ. Impact Assess. Rev. 77 (2019) 40–48, https://doi.org/10.1016/j.eiar.2019.03.007.
[82] Farmers for Climate Action, Renewable Energy in farming communities: summary report. https://farmersforclimateaction.org.au/wp-content/uploads/2025/02/FFCA-89DE-RenewableEnergyReport-2.pdf, 2024.
[83] KPMG, The human side of the energy transition. https://kpmg.com/content/dam/kpmg/au/pdf/2025/human-side-of-energy-transition-2025.pdf, 2025.
[84] EnergyCo, Community and Employment Benefit Program Consultation Feedback Report: Central-West Orana Renewable Energy Zone. https://www.energyco.nsw.gov.au/sites/default/files/2024-07/cebp-cwo-rez-consultation-feedback-report.pdf, 2024.
[85] Y. Cabannes, Contributions of participatory budgeting to climate change adaptation and mitigation: current local practices across the world and lessons from the field, Environ. Urban. 33 (2) (2021) 356–375, https://doi.org/10.1177/09562478211021710.
[86] J. Firestone, B. Hoen, J. Rand, D. Elliott, G. Hübner, J. Pohl, Reconsidering barriers to wind power projects: community engagement, developer transparency and place, J. Environ. Policy Plan. 20 (3) (2018) 370–386, https://doi.org/10.1080/1523908X.2017.1418656.
[87] K.W. Hoesch, S.B. Mills, J. Rand, R. Nilson, D.L. Bessette, J. White, B. Hoen, What to expect when you're expecting engagement: delivering procedural justice in large-scale solar energy deployment, Energy Res. Soc. Sci. 120 (2025) 103893, https://doi.org/10.1016/j.erss.2024.103893.
[88] J.A. Bingham, E. Miller, L. Ruscetta, H.A. Taufiq, G. Stelmach, J. Firestone, T. R. Johnson, S. Hazboun, H. Boudet, Understanding the roles, limits, and best practices for community benefit agreements in U.S. offshore wind development, Ocean Coast. Manag. 268 (2025) 107769, https://doi.org/10.1016/j.ocecoaman.2025.107769.
[89] C. Wright, D. Nyberg, V. Bowden, Beyond the discourse of denial: the reproduction of fossil fuel hegemony in Australia, Energy Res. Soc. Sci. 77 (2021) 102094, https://doi.org/10.1016/j.erss.2021.102094.
[90] D. Clarke, K. Baldwin, F. Baum, B. Godfrey, S. Richardson, L. Robin, T. Soliman Hunter, M. Taylor, Australian Energy Transition Research Plan. Report Four: Social Engagement Dynamics, Australian Council of Learned Academies, 2022. ttps://acola.org/wp-content/uploads/2022/05/ACOLA-2022-AETRP-Report-4-social-engagement-dynamics.pdf.